Reading the Letter, Not the Horizon: A Construct-Validity Audit of Temporal-Preference Probes

ANONYMOUS AUTHOR(S)

Linear probes on a language model's activations are proposed as oversight tools for decision-relevant distinctions that a model's final answer may not reveal. A probe's accuracy, however, is evidence about the intended construct only if nothing else in the data predicts the label. We audit probes for short- versus long-horizon choices, a distinction that bears on many normatively significant trade-offs, across the released training stages of OLMo 3 7B (Base, RL-Zero, SFT, DPO, Instruct) and in Qwen3-4B-Instruct. In the probe dataset we use, every short-horizon option is marked "(A)" and every long-horizon option "(B)". Probes trained on the original prompts reach 0.97–1.00 held-out accuracy, but swapping the two letters at test time reduces them to 0.03–0.11 in OLMo blocks 26–31, at every stage; a text classifier with no model reproduces the original scores. With the letters removed, a weaker signal survives: it transfers zero-shot to a second, explicit-horizon dataset at 0.85–0.87, against at most 0.60 for bag-of-words baselines. This signal is already present in Base; post-training leaves its peak transfer within 0.02 and lowers its final-block transfer by 0.04–0.06. Activation steering with it does not change forced choices more selectively than norm-matched random directions. We argue that representation-based oversight claims need artifact controls, discrimination from correlated constructs and causal tests, and report which of these our evidence meets.

Additional Key Words and Phrases: construct validity, linear probes, interpretability, dataset artifacts, intertemporal choice, oversight

1 Introduction

A model's final answer often under-determines the considerations behind it [27], which makes internal measurements attractive for oversight [5, 29]: a decision-relevant distinction read from activations could be tracked even when the output looks appropriate. Intertemporal trade-offs [9] are a natural target. Many decisions with ethical weight set an immediate benefit against delayed costs, obligations, or effects on people affected later, and how a system weighs them is rarely stated. A short horizon is not, by itself, the worse choice: containing harm now can be the right action. The construct of interest is therefore not "moral correctness" but whether and how a model represents the horizon of its options.

A probe is a classifier trained to predict a label from activations [1, 2]. High accuracy shows that the label is linearly decodable, not that the model encodes the intended construct, and still less that it uses it [8, 14, 21]. If some incidental property of the inputs predicts the label, a probe can learn that property instead. This is a construct-validity problem in the classical sense [6, 15], and the same one behavioral benchmarks face when a score reflects the format of the test rather than the competence it was built to measure [10, 12, 18].

We study it in a concrete case. A recent study localised a temporal-preference signal in Qwen3-4B-Instruct with probes, attribution and patching, using paired prompts that contrast an immediate and a long-term course of action [23]. We ask at which training stage such a signal forms, using the released checkpoints of OLMo 3 7B [25]: Base, RL-Zero (reinforcement learning applied directly to Base), SFT, DPO, and the final Instruct model (after RLVR).

The probe accuracies were near-perfect and transferred almost perfectly to a second dataset. Both facts have a simple cause: in both released datasets the short-horizon option is always labelled "(A)" and the long-horizon option always "(B)". Swapping the two letters while keeping the content fixed drives the late-layer probe far below chance; it predicts the letter. Removing the letters leaves a weaker signal that generalises across datasets better than any bag-of-words baseline we tried, is present in Base, and changes little through post-training. Steering along it does not selectively shift forced choices, and whether it encodes time horizon rather than a correlate such as the scope of the proposed action remains open.

Contributions. (1) A documented, model-independent label–format confound in a temporal-preference probe dataset, and a letter-swap test showing that it accounts for the late-layer probe in all five OLMo 3 stages and inflates it in Qwen3-4B (§3.1). (2) A letter-free re-analysis across training stages: a signal that transfers across datasets, is present before post-training, and is not shown to be causally used (§3.2–§3.4). (3) A graded account of the evidence a representation audit needs before it supports oversight claims, with the levels this study does and does not reach (§4).

2 Experimental design and validity controls

Data. Two publicly released sets [23]. The *implicit* set has 300 pairs: a stem and two continuations, one immediate and one long-term, with almost no temporal vocabulary (e.g. "When the system fails, the first priority becomes (A) containing the spreading damage before it escalates" versus "… (B) redesigning the architecture to prevent recurrence"). The *explicit* set has 500 pairs in the same format, with stated horizons ("… (A) What we can deliver in the next 30 days" versus "(B) What position we want in 10 years"). Each continuation is a separate prompt containing *one* option, labelled 1 if long-term, giving 600 and 1,000 prompts. The option letter determines the label in every prompt of both sets (Appendix A).

Models and formats. The five OLMo 3 7B checkpoints above, 32 decoder blocks, run in bfloat16. Base has no chat template, so every checkpoint is run on the raw prompt (*plain*); the post-trained checkpoints are also run inside their chat template (*chat*). As a cross-model check we run Qwen3-4B-Instruct-2507 (36 blocks, chat), the model of the original study. Our probe and readout follow our own pipeline, so this is a check of the dataset, not a reproduction of that study.

Probe. We read the residual stream at the output of every block at the last token of the option text (in chat format, the last token of the user turn, before the template's end-of-turn tokens). Features are standardised with training-split statistics, and an ℓ_2-regularised logistic regression (C=0.1) is fitted with L-BFGS in float64. Splits are 80/20 and grouped by pair, so both continuations of a pair are on the same side; we use five such splits, the same five for every checkpoint. *Implicit accuracy* is measured on the 60 held-out pairs (120 prompts); *explicit transfer* applies the implicit-trained probe, unchanged, to all 1,000 explicit prompts. Reported spread is the standard deviation over the five splits, which varies the training data; the explicit test set is fixed, so it does not include test-sampling uncertainty.

Controls. Letter swap: "(A)" and "(B)" are exchanged in each prompt, content unchanged. *Letter removal*: the markers are deleted. The *original-format probe* is trained on the data as released and evaluated under each condition; the *letter-free probe* is trained and evaluated without letters. *Lexical baselines* are bag-of-words logistic regressions with no model: count and TF-IDF features over unigrams or unigrams+bigrams, four regularisation strengths, on the same splits. We report the best of the 12 settings chosen on the evaluation data, which favours the baseline and so makes the comparison conservative.

3 Results

3.1 A near-perfect probe fails letter counterbalancing

Trained on the released prompts, the probe for the Instruct model (chat format) reaches 0.97–1.00 on held-out implicit pairs from block 7 on and transfers at 0.97 to the explicit set at block 26 (Fig. 1). Because the two sets share little vocabulary (Jaccard similarity 0.21; Appendix A), that transfer would ordinarily count against a lexical explanation. Swapping the letters at test leaves early blocks close to their original accuracy, drops the probe below chance from block 17, and leaves it at 0.04–0.06 on implicit pairs in blocks 26–31 and 0.11–0.25 on the explicit set in blocks 20–31.

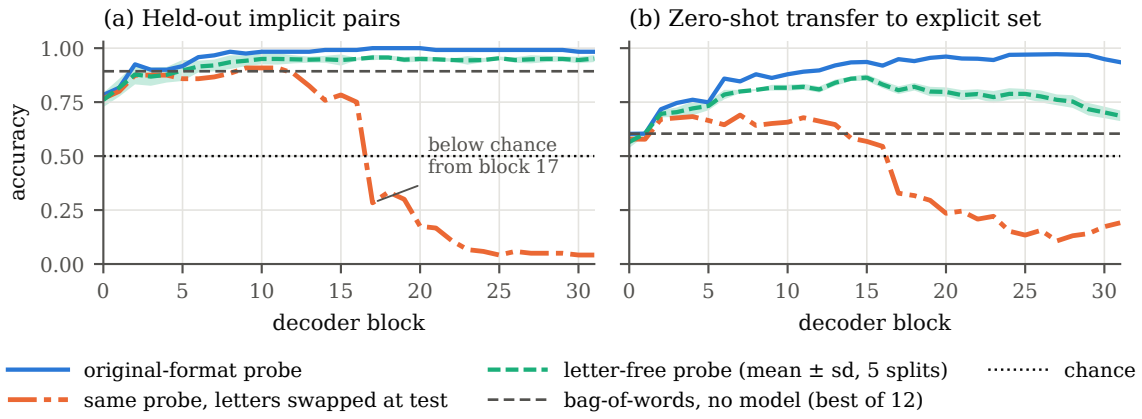


Fig. 1. Option-letter confound, OLMo 3 Instruct, chat format. (a) Held-out implicit pairs; (b) the same probes applied zero-shot to the explicit set. The original-format probe (solid) is near-perfect, but evaluating it with (A)/(B) swapped (dash-dot) drops it below chance from block 17. A probe trained without letters (dashed; band: ±1 sd over five grouped splits) keeps a weaker signal. Grey dashed: best letter-free bag-of-words baseline; dotted: chance.

Far-below-chance accuracy means the probe's decision follows the letter, not that the activations contain nothing else: the same layers support a good letter-free probe (§3.2).

The pattern holds at every stage and in both formats: under the swap, the original-format probe scores 0.03–0.11 in blocks 26–31 for Base, RL-Zero, SFT, DPO and Instruct (Table 1, last column). It is a property of the dataset and readout, and needs no model at all: a bag-of-words classifier that keeps the letter tokens scores 1.00 on implicit pairs and 0.99 on the explicit set. Qwen3-4B-Instruct shows the same failure with a different depth profile. At layer 26, where the original study's probe peaked, our original-format probe scores 0.99 but 0.29 with letters swapped, falling to 0.03 in layers 32–35. Unlike in OLMo, a letter-free probe at layer 26 remains strong (0.95 implicit, 0.87 explicit), so in Qwen the confound inflates the late-layer probe rather than accounting for all of it.

3.2 A letter-free signal survives but is only partly identified

With the letters removed, the probe separates held-out implicit pairs at 0.95–0.97, only modestly above the best bag-of-words baseline (0.89 ± 0.03): within the implicit set, the words are nearly sufficient. Transfer is more informative. Lexical baselines trained on implicit items reach at most 0.60 on the explicit set, while the letter-free probe transfers at 0.85–0.87 at its best block in every OLMo checkpoint (Table 1). The probe therefore captures something shared across the two sets that word identity does not.

What that shared property is, is less clear. The implicit set contrasts *courses of action* (contain the damage versus redesign the architecture) and almost never states a horizon (1% of prompts contain a time expression from our list, versus 42% of short and 66% of long options in the explicit set). A direction that separates the implicit options could encode time horizon, but equally the scope, abstractness or ambition of the action, all of which co-vary with horizon in both sets. The two sets also come from the same study and share their template, so transfer between them is weaker evidence of generality than transfer to an independently constructed set would be. We therefore call the signal *horizon-correlated*: robust to the letter artifact and to lexical overlap, not yet discriminated from correlated constructs.

Table 1. Letter-free probe by checkpoint (*Implicit*: held-out accuracy at the best block; *Explicit*: zero-shot transfer at the best block, given in *Best block*, and at the final block), and the original-format probe evaluated with letters swapped. Means over five grouped splits; split-to-split sd ≤0.02 at the best block and ≤0.03 at the final block. Best letter-free bag-of-words baselines: 0.89 implicit, 0.60 explicit. Qwen3-4B has 36 blocks; its final block is 35.

		Letter-free probe				Original probe,
Model	Checkpoint, format	Implicit	Explicit	Best block	Explicit, block 31	swapped, block 26
OLMo 3 7B	Base, plain	0.97	0.86	10	0.78	0.05
	RL-Zero, plain	0.97	0.86	10	0.78	0.04
	SFT, plain	0.96	0.86	15	0.74	0.11
	DPO, plain	0.96	0.87	15	0.73	0.11
	Instruct, plain	0.97	0.87	15	0.72	0.08
	SFT, chat	0.96	0.85	15	0.71	0.08
	DPO, chat	0.96	0.86	15	0.69	0.07
	Instruct, chat	0.96	0.86	15	0.69	0.06
Qwen3-4B	Instruct, chat (cross-model check)	0.97	0.89	21	0.82	0.29

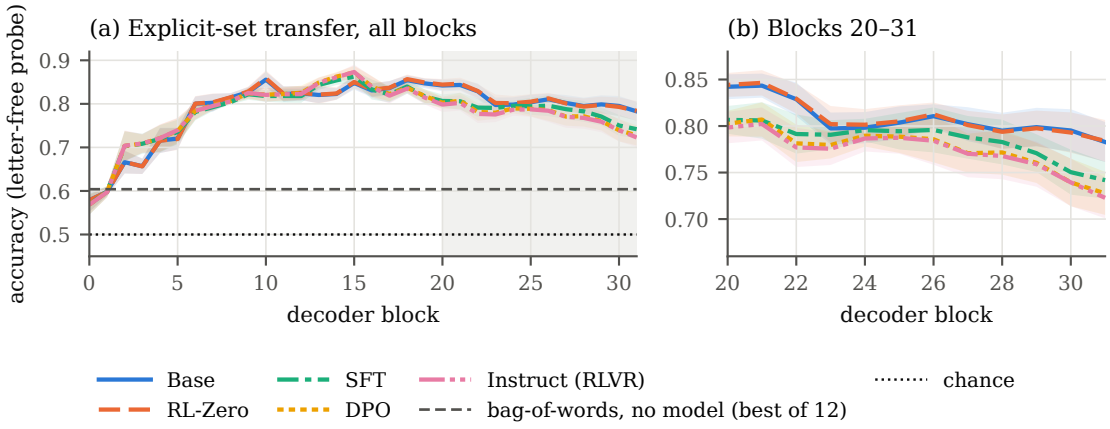


Fig. 2. Letter-free probe, zero-shot transfer to the explicit set, OLMo 3 checkpoints in plain-text format (mean ±1 sd over five grouped splits). (a) All blocks; (b) blocks 20–31 enlarged. Base and RL-Zero coincide (the RL-Zero curve hides Base). From SFT on, transfer is lower in the final blocks; the peak is unchanged.

3.3 Post-training changes late-layer readout more than peak transfer

All OLMo checkpoints peak at 0.85–0.87, within 0.02 of one another, and their curves overlap up to block 20 (Fig. 2). The best block shifts from 10 (Base, RL-Zero) to 15 (SFT onwards), but the curves are flat between blocks 10 and 20, so we do not treat the shift as a difference. The final blocks do differ. Comparing each checkpoint with Base on the same split, block-31 transfer is lower for SFT by 0.041 on average (range 0.030–0.051, lower in all five splits), with smaller further decreases for DPO (0.014 beyond SFT) and Instruct (0.005 beyond DPO), again in all five splits. In chat format the final-block values are lower still (0.69–0.71). RL-Zero and Base differ by at most 0.007 at any block in any split.

Three cautions apply. The checkpoints are observational snapshots: SFT differs from Base in data, objective and duration, so we can locate the change but not attribute it to a mechanism. Five splits that vary only the training data are a narrow basis for uncertainty; we have not resampled evaluation items.

Finally, near-identical curves for Base and RL-Zero mean this measurement does not distinguish them, not that RL-Zero left the representation unchanged. Within these limits, the signal is present before post-training, which mainly lowers its linear readability in the last blocks.

3.4 Steering does not establish selective preference control

We add $k\,d$ to every position at the output of block $L \in \{8, 12, 16, 20, 26\}$ of the Instruct model while it answers an explicit-set pair as a forced choice ("Answer with just the letter A or B"), asked with both letter orders. d is the class-mean difference of letter-free chat-format activations on the implicit set; a letter-confounded direction (same difference on the original prompts) and a Gaussian random direction are rescaled to $\|d\|$, so k is in units of class separation. We report the next-token probabilities of "A" and "B", both conditional and summed (A+B mass).

Unsteered, the model chooses the long-horizon option with mean probability 0.50 (A+B mass 0.999). The largest increase is 0.10 (k=+2, blocks 8 and 12), and the letter-confounded direction produces the same increase; at block 16 the random direction moves the choice as much as the letter-free one (0.58 for both); at blocks 20 and 26 no direction moves it by more than 0.02 (Fig. 3). What changes more is the letter: among interventions that keep A+B mass ≥0.9, P("B") ranges from 0.17 to 0.82, and the random direction moves it as much as the others. At block 8, for example, the letter-free and letter-confounded directions lower P("B") at k=+2 (0.20 and 0.27) and raise it at k=−4 (0.66 and 0.82), while the random direction lowers it to 0.17 at block 8, k=−4, and raises it to 0.80 at block 16, k=+4. The share of prompts preferring the long-horizon option is 0.49 unsteered and at most 0.63. At $k = \pm 4$ in blocks 8–12 the model often stops answering with a letter (A+B mass down to 0.007; Appendix C). With aggregate means only, not per-prompt outcomes, we cannot test whether the 0.10 shifts differ from the random control. We read this as no evidence of selective control under this readout, not as evidence that the representation is unused: the counterbalanced forced choice averages to exactly indifference, and may not elicit this model's preference at all.

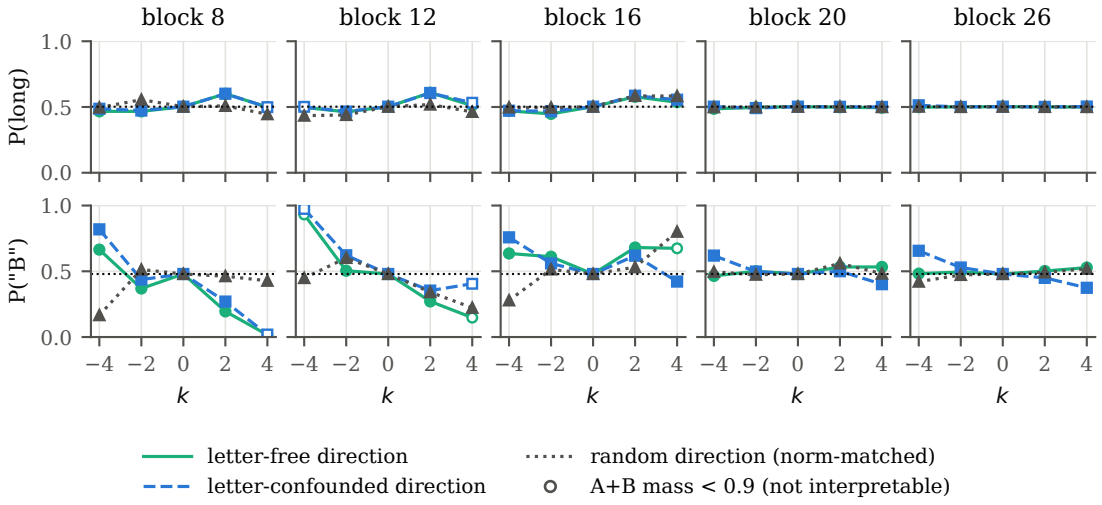


Fig. 3. Steering OLMo 3 Instruct at one block with strength k (A/B forced choice on the explicit set, both letter orders, 1,000 prompts). Top: P(long-horizon option); bottom: P("B"); both conditional on answering "A" or "B". Hollow markers: A+B mass <0.9, not interpreted. Dotted: unsteered.

Table 2. A graded view of the evidence a probe-based audit can provide, and what this study shows at each level. Proposed as an organising framework, not an established standard.

Level	Question	This study
1. Predictive validity	Does the probe predict the label on held-out items?	Yes (0.97–1.00), but satisfied by the letter alone.
2. Artifact robustness	Does it survive counterbalancing of irrelevant format features?	Original probe: no. Letter-free probe: yes, by construction.
3. Construct discrimination	Does it separate the construct from plausible correlates?	Not tested; horizon co-varies with action scope and abstractness.
4. Generalisation	Does it transfer across datasets and contexts?	Partly: 0.85–0.87 to a second set from the same study.
5. Causal relevance	Does intervening on it change behaviour selectively?	Not shown under the A/B readout.
6. Oversight utility	Does it predict decision-relevant failures better than simpler monitors?	Not tested.

4 Implications for representation-based oversight

High accuracy can be evidence for the wrong construct. Telling sound reasoning from merely appropriate-looking answers matters for representation audits too: a probe can score near-perfectly while reading the answer format.

Table 2 separates the claims a probe result can support. The original probe passes Level 1 and fails Level 2, and a failure there removes the support for every later claim, since a monitor built on it would track the letter. The letter-free probe passes Level 2 and partly Level 4. Counterbalancing is necessary but not sufficient: it removes one known artifact and is silent about unknown ones and about correlated constructs (Level 3), which is where the open question about this signal lies.

Decodable is not deployed. Levels 1–4 concern what is linearly accessible in the activations; Level 5 concerns what the model uses to choose [8, 21]. Even causal-looking evidence can mislead [17]. Our results separate the two: a horizon-correlated distinction is accessible from mid-network on, in every checkpoint, while steering along it does not selectively move choices in the tested format. An oversight claim of the form "the model is weighing the long-term consequences" needs the second kind of evidence, not only the first.

Value-laden contrasts need neutral operationalisations. Value-laden evaluations often pose paired or multiple-choice options [13, 24]; pairing a "patient" or "responsible" option with its opposite invites fixed conventions, such as listing the favoured option second, and behavioural scores and probes can then reflect position. Letter-swap and letter-removal controls cost one extra forward pass per prompt. The labels also need care: the immediate option is sometimes the better one, so a representation of horizon is not one of good judgement.

Where intervention can act. If the decodable signal forms before post-training and post-training mainly changes its late-layer readout, post-training is a weak lever on the representation itself. Resting on one model family and an unvalidated construct, this is a hypothesis for controlled training experiments, not a conclusion.

A minimal reporting standard. With any probe offered as evidence about a value-relevant disposition, report a format control, the strongest lexical baseline, transfer to an independent set, a matched contrast for the likeliest correlate, and a causal test against random directions. This study meets the first two, the third only partly, and not yet the last two.

5 Related work

Probing methodology has long distinguished decodability from use: control tasks and selectivity [14], the gap between probe accuracy and task relevance [21], and amnesic interventions that test whether removing a property changes behaviour [8]; see [2] for a review. Our confound is a dataset artifact in the sense of [12, 18, 20] and an instance of shortcut learning [10]; language models are also known to be sensitive to option letters and order behaviourally [19, 28].

Measurement-validity frameworks [6, 15] separate a construct from its operationalisation, and underspecification [7] explains why equal held-out scores can rest on different features. Time is linearly represented in language-model activations [11]; activation addition [16, 22, 26] is our causal test. Training-stage analyses have mostly used pretraining checkpoints [4]; OLMo 3 releases post-training stages as well [25]. The temporal-preference study we audit [23] used attribution and patching in addition to probes; our check concerns the dataset and our own probe, not those analyses.

6 Limitations and conclusion

Each limitation below names the experiment that would address it. *Construct*: horizon is confounded with action scope and abstractness. A proposed, not yet run, test of Level 3 crosses horizon (short, long) with action scope (local fix, structural change) at matched stems, lengths and counterbalanced order, plus a non-temporal concrete-versus-abstract contrast at fixed horizon. The letter-free probe separating horizon at fixed scope, and not scope or abstractness at fixed horizon, would support a horizon reading; separating scope or abstractness as well would count against it. *Data independence*: both sets share one study and template; transfer to an independently built set with other option formats would test generality. *Models*: one family for stages, one cross-model check; another family with released post-training stages would test whether the SFT effect is general. *Training attribution*: released snapshots cannot isolate a mechanism; probing intermediate pretraining checkpoints and controlled fine-tuning runs from Base would. *Causal readout*: the A/B readout averages to indifference; free-text continuations scored for horizon, with per-prompt outcomes for a paired test against random directions, and concept erasure [3] would allow a real causal test. *Procedure*: one token position and one linear probe; results may depend on position, format and regularisation.

A strong internal score was, in this case, evidence for the wrong construct. A one-line counterfactual control revealed it, and the signal that survives is real, robust to the artifact and present from pretraining on, but not yet shown to be about time or to drive choices. Before probes of value-relevant distinctions are used as oversight evidence, they should be shown to survive format counterbalancing, to discriminate the intended construct from its correlates, and to matter causally.

A Dataset artifact audit

Transformations. Letter swap exchanges "(A)" and "(B)"; with one marker per prompt, this flips the option's letter ("… becomes (B) redesigning …" → "… becomes (A) redesigning …"). Letter removal deletes the marker and surrounding whitespace.

Table 3. Option letter by label in the released datasets. Each prompt contains one option.

	Implicit (300 pairs)		Explicit (500 pairs)	
	"(A)"	"(B)"	"(A)"	"(B)"
Short-horizon (label 0)	300	0	500	0
Long-horizon (label 1)	0	300	0	500

Lexical overlap and time expressions. Lower-cased word vocabularies (letters removed) have 1,727 types (implicit) and 1,200 (explicit), 511 shared (Jaccard 0.21); 44% of explicit word tokens occur in the implicit vocabulary. A fixed list of time expressions (*day(s), week(s), month(s), year(s), decade(s), hour(s), minute(s), quarter, century, generation(s), future, long-term, short-term, immediate(ly), now, today, tomorrow, soon, later*) matches 1% of implicit prompts in either class, and 42% (short) and 66% (long) of explicit prompts.

B Protocol details and reproducibility

Split 0 is a fixed grouped 80/20 split (240/60 pairs); splits 1–4 sample 60 test pairs without replacement with NumPy seeds 1–4. The same five splits are used for every checkpoint and for the baselines. The probe standardises each feature with training-split mean and standard deviation, then minimises $C \sum_i \ell_{\mathrm{BCE}}(w^\top z_i + b, y_i) + \frac{1}{2}\|w\|^2$ with C=0.1 (L-BFGS, at most 500 iterations, float64); on one checkpoint it matched a scikit-learn fit to within one test example. Activations are cached in float16. The baseline grid is {binary counts over unigrams, binary counts over unigrams+bigrams, TF-IDF over unigrams+bigrams} $\times$ $C \in \{0.1, 1, 10, 100\}$ (scikit-learn logistic regression); without letters the best implicit accuracy is 0.893 ± 0.028 and the best explicit transfer 0.604 ± 0.007 (both TF-IDF); with letters every setting scores 1.00 implicit and 0.990–0.996 explicit.

Paired stage differences. Block-31 explicit transfer (letter-free, plain), difference to the previous stage on the same split, mean (range): DPO−SFT −0.014 (−0.008 to −0.025); Instruct−DPO −0.005 (−0.003 to −0.009); RL-Zero−Base +0.001 (−0.003 to +0.004). Averaged over blocks 20–31, SFT−Base is −0.024.

Reproducibility. Checkpoints (Hugging Face): allenai/Olmo-3-1025-7B at revision a81bae4; allenai/Olmo-3-7B-RL-Zero-General; allenai/Olmo-3-7B-Instruct-SFT; allenai/Olmo-3-7B-Instruct-DPO; allenai/Olmo-3-7B-Instruct; Qwen/Qwen3-4B-Instruct-2507. All but Base were used at the default revision when downloaded (October 2026). Inference in bfloat16 with Hugging Face Transformers on single 24 GB GPUs (under an hour for all six models). Exact package versions were not recorded for all runs. We will release code and per-layer results with the de-anonymised version.

C Steering diagnostics

Table 4. Steering at k=+2 ($k \in \{\pm 2, \pm 4\}$ were run; direction added at every position of the block output): P(long-horizon option) / P("B"), conditional on "A" or "B". Unsteered: 0.50 / 0.48, A+B mass 0.999.

Block	Letter-free	Letter-confounded	Random
8	0.60 / 0.20	0.60 / 0.27	0.51 / 0.46
12	0.61 / 0.27	0.61 / 0.35	0.52 / 0.34
16	0.58 / 0.68	0.59 / 0.62	0.58 / 0.53
20	0.50 / 0.53	0.50 / 0.50	0.50 / 0.56
26	0.50 / 0.50	0.50 / 0.45	0.50 / 0.49

Raw A+B mass. (before renormalising) is below 0.9 only at block 8, k=+4 (letter-free 0.007, letter-confounded 0.016), block 12, k= ± 4 (0.32 and 0.18; 0.10 and 0.22) and block 16, k=+4 (letter-free 0.88); the random direction (one Gaussian draw per block, seed 0) never takes it below 0.96.

References

[1] Guillaume Alain and Yoshua Bengio. 2017. Understanding Intermediate Layers Using Linear Classifier Probes. In *International Conference on Learning Representations, Workshop Track*.

[2] Yonatan Belinkov. 2022. Probing Classifiers: Promises, Shortcomings, and Advances. *Computational Linguistics* 48, 1 (2022), 207–219.

[3] Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. 2023. LEACE: Perfect Linear Concept Erasure in Closed Form. In *Advances in Neural Information Processing Systems*.

[4] Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. In *International Conference on Machine Learning*.

[5] Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2023. Discovering Latent Knowledge in Language Models Without Supervision. In *International Conference on Learning Representations*.

[6] Lee J. Cronbach and Paul E. Meehl. 1955. Construct Validity in Psychological Tests. *Psychological Bulletin* 52, 4 (1955), 281–302.

[7] Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. 2022. Underspecification Presents Challenges for Credibility in Modern Machine Learning. *Journal of Machine Learning Research* 23, 226 (2022), 1–61.

[8] Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. 2021. Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals. *Transactions of the Association for Computational Linguistics* 9 (2021), 160–175.

[9] Shane Frederick, George Loewenstein, and Ted O'Donoghue. 2002. Time Discounting and Time Preference: A Critical Review. *Journal of Economic Literature* 40, 2 (2002), 351–401.

[10] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut Learning in Deep Neural Networks. *Nature Machine Intelligence* 2 (2020), 665–673.

[11] Wes Gurnee and Max Tegmark. 2024. Language Models Represent Space and Time. In *International Conference on Learning Representations*.

[12] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation Artifacts in Natural Language Inference Data. In *Proceedings of NAACL-HLT*.

[13] Dan Hendrycks, Collin Burns, Steven Basart, Andrew Critch, Jerry Li, Dawn Song, and Jacob Steinhardt. 2021. Aligning AI With Shared Human Values. In *International Conference on Learning Representations*.

[14] John Hewitt and Percy Liang. 2019. Designing and Interpreting Probes with Control Tasks. In *Proceedings of EMNLP-IJCNLP*.

[15] Abigail Z. Jacobs and Hanna Wallach. 2021. Measurement and Fairness. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 375–385.

[16] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-Time Intervention: Eliciting Truthful Answers from a Language Model. In *Advances in Neural Information Processing Systems*.

[17] Aleksandar Makelov, Georg Lange, and Neel Nanda. 2023. Is This the Subspace You Are Looking For? An Interpretability Illusion for Subspace Activation Patching. *arXiv preprint arXiv:2311.17030* (2023).

[18] R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the Wrong Reasons: Diagnosing Syntactic Heuristics in Natural Language Inference. In *Proceedings of ACL*.

[19] Pouya Pezeshkpour and Estevam Hruschka. 2024. Large Language Models Sensitivity to The Order of Options in Multiple-Choice Questions. In *Findings of NAACL*.

[20] Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. Hypothesis Only Baselines in Natural Language Inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics (*SEM)*.

[21] Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. 2021. Probing the Probing Paradigm: Does Probing Accuracy Entail Task Relevance?. In *Proceedings of EACL*.

[22] Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2024. Steering Llama 2 via Contrastive Activation Addition. In *Proceedings of ACL*.

[23] Ian Rios-Sialer, Shantanu Darveshi, Shuai Jiang, Avigya Paudel, Anastasiia Pronina, Ipshita Bandyopadhyay, and Justin Shenk. 2026. Temporal Preference Concepts and their Functions in a Large Language Model. *arXiv preprint arXiv:2606.05194* (2026).

[24] Nino Scherrer, Claudia Shi, Amir Feder, and David M. Blei. 2023. Evaluating the Moral Beliefs Encoded in LLMs. In *Advances in Neural Information Processing Systems*.

[25] Team OLMo. 2025. Olmo 3. *arXiv preprint arXiv:2512.13961* (2025).

[26] Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. 2023. Steering Language Models With Activation Engineering. *arXiv preprint arXiv:2308.10248* (2023).

[27] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. In *Advances in Neural Information Processing Systems*.

[28] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2024. Large Language Models Are Not Robust Multiple Choice Selectors. In *International Conference on Learning Representations*.

[29] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. 2023. Representation Engineering: A Top-Down Approach to AI Transparency. *arXiv preprint arXiv:2310.01405* (2023).